

Deep Learning Approaches for Pneumonia Detection: Multi-Scale Analysis with Explainable AI

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Abstract—This study presents a comprehensive analysis of CNN architecture fine-tuning across multiple input resolutions combined with systematic explainable AI (XAI) evaluation for pneumonia detection in chest X-rays. We fine-tuned four CNN architectures (AlexNet, ResNet-18, ResNet-50, and DenseNet-169) from 28×28 to higher resolutions (64×64 and 128×128) using the PneumoniaMNIST dataset and evaluated their performance alongside three XAI techniques (Grad-CAM, LIME, and SHAP). Our findings reveal a critical tension between sensitivity and overall performance: while ResNet-50 and AlexNet achieved zero false negatives at 28×28 resolution with data augmentation, fine-tuning to higher resolutions improved overall accuracy (ResNet-18: 0.880 → 0.907) but increased false negatives. XAI analysis demonstrated that higher resolutions (128×128) significantly improve explanation quality and anatomical correspondence, with Grad-CAM proving most robust across architectures. Our systematic evaluation reveals that data augmentation at low resolution can be more effective than resolution scaling for critical error reduction, and that architectural design fundamentally determines a model’s ability to benefit from higher-resolution inputs. These findings provide evidence-based guidelines for medical AI deployment: ResNet-50 at 28×28 with augmentation for maximum sensitivity screening, and ResNet-18 at 128×128 for balanced diagnostic support with superior interpretability.

Index Terms—Deep Learning, Medical Image Classification, Pneumonia Detection, False Negative Rate Optimization, ResNet, AlexNet, DenseNet, Data Augmentation, Transfer Learning, Fine-tuning, Chest X-ray Analysis, Explainable AI (XAI), Grad-CAM, LIME, SHAP.

I. INTRODUCTION

This project builds upon our previous work on the automated analysis of chest X-rays for the detection of pneumonia. While the initial study focused on evaluating deep learning classification models under various hyperparameter configurations (such as batch size, dropout, and data augmentation), this new phase takes us closer to achieving a clinically viable and transparent solution by integrating explainable artificial intelligence (XAI) techniques into the model evaluation process.

I-A BACKGROUND

Deep learning has transformed medical image analysis, delivering exceptional performance in disease detection tasks

such as pneumonia classification from chest X-rays. However, despite their accuracy, many state-of-the-art models operate as ‘black boxes’, posing a critical challenge to their adoption in healthcare environments, where transparency, trust, and clinical validation are essential.

Explainable AI (XAI) tools, such as Grad-CAM, LIME, and SHAP, have therefore emerged as promising approaches to enhance interpretability. These methods provide visual or feature-level explanations to help clinicians understand and verify the model’s decisions [3], [4].

While many recent lightweight architectures such as MobileNet and EfficientNet have been proposed to address computational constraints in clinical settings [2], [4], our study focuses on widely adopted convolutional models—AlexNet, ResNet-18, ResNet-50, and DenseNet-169—due to their proven effectiveness in medical imaging. These models also provide a strong baseline for investigating how interpretability varies across architectures of different depths and complexities.

In this follow-up study, we extend our initial experiments by fine-tuning these models at higher input resolutions (from 28×28 to 64×64 and 128×128), and systematically analyzing the interpretability of their predictions using multiple XAI techniques.

I-B PROBLEM STATEMENT AND MOTIVATION

Although convolutional neural networks (CNNs) have demonstrated high classification performance on medical imaging tasks, several limitations hinder their practical deployment:

- Lack of interpretability in deep learning predictions.
- Limited validation of explainability tools in clinical workflows.
- Uncertainty about how different XAI methods behave across CNN architectures.
- The effect of image resolution on both classification performance and explanation quality remains underexplored.

This project addresses these limitations by comparing saliency-based and model-agnostic XAI techniques across

multiple pre-trained CNN models, with attention to how these explanations change when models are fine-tuned to different image resolutions.

I-C OBJECTIVES AND CONTRIBUTIONS

This study investigates the use of explainable deep learning for detecting pneumonia in chest X-ray images, as an extension of our earlier model benchmarking. Our contributions are as follows:

- 1) Fine-tune and evaluate previously tested CNN architectures (AlexNet, ResNet-18, ResNet-50, DenseNet-169) at higher resolutions (64x64 and 128x128) using insights from prior hyperparameter tuning.
- 2) Apply and compare three prominent XAI techniques—Grad-CAM, LIME, and SHAP—to visualize and explain model predictions.
- 3) Analyze how different XAI methods behave across model architectures and resolutions, both qualitatively (visual overlays) and quantitatively (explanation coverage and consistency).
- 4) Provide insights into the trade-offs between model accuracy, complexity, and interpretability in the context of pneumonia detection.

I-D ORGANIZATION

This paper is organized as follows:

- Section I presents the background, motivation, and contributions.
- Section II reviews related literature on medical image classification and explainable AI.
- Section III provides theoretical foundations for the XAI techniques applied.
- Section IV describes the dataset and preprocessing pipeline.
- Section V details the experimental design, models, training setup, and XAI implementation.
- Section VI presents the classification and interpretability results, with visualizations and metrics.
- Section VII summarizes the findings and outlines future work directions.

II. RELATED WORK

This section reviews existing literature on deep learning models for pneumonia detection using chest X-rays, with emphasis on explainable AI (XAI) integration. We examine standardized datasets, architectural developments, interpretability tools, and gaps that our study seeks to address.

II-A MEDICAL IMAGE CLASSIFICATION FOR PNEUMONIA DETECTION

Firstly, we survey the datasets and standardization benchmarks that establish pneumonia detection research. We then review leading model architectures - both classic CNNs and recent

transformer-based approaches - and conclude by identifying remaining methodological and clinical gaps.

II-A1 Datasets and Standardization Efforts

A recurring challenge in the analysis of medical images for the detection of pneumonia is the lack of standardized datasets and processing pipelines. Although their work is not exclusively focused on pneumonia, van der Velden et al. (2022) [2] emphasize the importance of standardization in deep learning workflows when integrating explainable artificial intelligence (XAI) into medical image analysis. This highlights the necessity for consistency in data pre-processing and annotation when developing interpretable models more broadly.

Conversely, Novokshanov (2024) [6] presents a case study targeting pneumonia detection specifically, which demonstrates how real world datasets can be used alongside XAI to deliver transparent diagnostic insights. His work indirectly reflects efforts to structure datasets through its practical application.

II-A2 Model Development and Approaches

Recent contributions have showcased practical approaches to building pneumonia classification models using explainable mechanisms. For example, Sheu et al. (2023) [5] have proposed a specific pneumonia classification pipeline incorporating CNN-based architectures and XAI techniques such as SHAP and Grad-CAM to highlight relevant regions of images.

Similarly, Novokshanov (2024) [6] demonstrates the feasibility of real-time localization of pneumonia regions using saliency maps, reinforcing the potential for lightweight, interpretable tools in clinical settings.

II-A3 Methodology Refinement

Advancements in model interpretability have prompted researchers to refine traditional classification methodologies. Borys et al. (2023) [4] provide a detailed review of saliency-based XAI methods such as Grad-CAM, offering valuable insights into how these tools can enhance the visual interpretability of CNN outputs. Their analysis emphasizes the importance of selecting XAI techniques that are appropriate for the medical context.

Similarly, van der Velden et al. (2022) [2] discuss how XAI can be coherently integrated into clinical workflows, contributing to best practice in methodology refinement by aligning algorithmic outputs with medical professionals' expectations and understanding.

II-B PRE-TRAINED MODELS IN MEDICAL IMAGE CLASSIFICATION

Pre-trained convolutional neural networks (CNNs) remain foundational in medical image analysis due to their hierarchical feature extraction capabilities. However, integrating transparency into their decision-making is critical for clinical application, as these models often function as "black boxes."

As a result, many recent studies have augmented pre-trained CNNs with explainability methods such as Grad-CAM, LIME, and SHAP to improve interpretability and trust.

II-B1 Grad-CAM (Gradient-weighted Class Activation Mapping)

Grad-CAM is one of the most widely used saliency methods, providing pixel or region level explanations. Sheu et al. (2023) [5] incorporate Grad-CAM in their pneumonia classification pipeline to highlight areas influencing predictions. Novoksharov (2024) [6] also uses it for localization in Pneumonia-MNIST data, reinforcing Grad-CAM’s utility in identifying disease-specific regions.

II-B2 LIME and SHAP techniques

In addition to Grad-CAM, model-agnostic techniques such as LIME (Local Interpretable Model-Agnostic Explanations) and SHAP (SHapley Additive Explanations) provide further transparency, particularly in complex or ensemble-based architectures. While these methods were not the primary focus of the reviewed articles, their potential was acknowledged in broader discussions, such as those by Borys et al. (2023) [4], who emphasized the need for explanations that are clinically meaningful rather than purely technical. LIME and SHAP are particularly useful when adapting existing models to new datasets or using pre-trained models, as they do not require the model to be retrained to generate interpretable outputs.

In addition to interpretability, computational efficiency is an increasing concern, particularly in healthcare settings with limited resources. While most of the reviewed studies primarily utilize standard CNN architectures, recent literature increasingly explores the integration of lightweight models such as EfficientNet and MobileNet. These architectures offer a favorable balance between accuracy and computational cost, making them well suited to deployment on mobile devices or in clinical settings with limited infrastructure. For instance, van der Velden et al. (2022) [2] and Borys et al. (2023) [4] both discuss the necessity of efficient, interpretable models, even though they do not explicitly implement these architectures. Their observations emphasize the importance of moving towards models that are powerful, efficient and explainable.

II-B3 Summary

Pre-trained CNNs augmented with XAI tools—especially Grad-CAM, LIME, and SHAP—provide varying levels of interpretability. However, challenges remain in comparing their impact across architectures and resolutions, as well as validating them clinically. This motivates our investigation into how XAI techniques perform across different CNN depths and image resolutions to inform practical deployment in pneumonia diagnosis.

II-C GAP ANALYSIS IN CURRENT APPROACHES

II-C1 Evaluation Limitations

Evaluation practices in explainable medical imaging are still limited by an overemphasis on quantitative metrics and a lack of clinical context. Abrantes and Rouzrokh (2024) [3] offer a critical perspective on the limitations of current interpretability evaluations, contending that existing metrics frequently overlook the reliability of explanations from a clinical standpoint.

Similarly, van der Velden et al. (2022) [2] highlight the technical limitations of popular XAI methods, such as their susceptibility to noise and perturbations. These limitations emphasize the urgent need for evaluation frameworks that consider both technical performance and clinical relevance.

II-C2 Methodological Gaps

Although saliency-based methods are widely used, they are often misunderstood or misapplied in medical contexts. Borys et al. (2023) [4] highlight this issue, noting that many practitioners adopt these tools without grasping their underlying principles or limitations.

Furthermore, Abrantes and Rouzrokh (2024) [3] argue that there is still no consensus on what constitutes a ‘good’ explanation, reflecting a deeper methodological gap that impedes progress in clinically relevant XAI. These observations suggest that, as technology develops, the rigor of the methodology must also improve.

II-C3 Clinical Integration Challenges

One of the most persistent barriers to adopting XAI for pneumonia detection is the clinical integration of these tools. Both van der Velden et al. (2022) [2] and Abrantes and Rouzrokh (2024) [3] discuss the practical challenges involved in incorporating XAI solutions into routine clinical workflows. Issues such as a lack of validation in real-world settings, limited trust from medical professionals and misalignment with workflows are central obstacles.

Borys et al. (2023) [4] also emphasize the importance of developing XAI tools that are technically sound and designed with clinicians in mind. Without such considerations, even the most accurate and explainable models risk being underutilized or misunderstood in practice.

This study aims to address these gaps by:

- Fine-tuning standardized CNN models (AlexNet, ResNet-18/50, DenseNet-169) at multiple image resolutions (64x64 and 128x128), building on 28x28 baselines.
- Applying Grad-CAM, LIME, and SHAP to each model/resolution configuration.
- Conducting comparative analysis of visual explanation quality, consistency, and clinical relevance across architectures and resolutions.

III. EXPLAINABLE AI TECHNIQUES

This section provides the theoretical foundation for the three XAI techniques applied in our study: Grad-CAM, LIME, and

SHAP. Understanding their underlying principles is essential for interpreting the results presented in Section VI.

III-A GRAD-CAM (GRADIENT-WEIGHTED CLASS ACTIVATION MAPPING)

Grad-CAM (Gradient-weighted Class Activation Mapping) is a method specific to models that provides local explanations for deep neural networks. As an explainability technique, it visually highlights the regions of an image that are most important for the network’s classification decision.

It computes the gradients of the model’s output with respect to the feature maps in the final convolutional layer and reveals which parts of the image the model ‘looks at’ when making a prediction [12]. These gradients are globally averaged to obtain importance weights, which are then combined with the feature maps to create a heatmap showing which parts of the image contributed most strongly to the decision [13]. The computational steps for generating Grad-CAM heatmaps are illustrated in Figure 1.

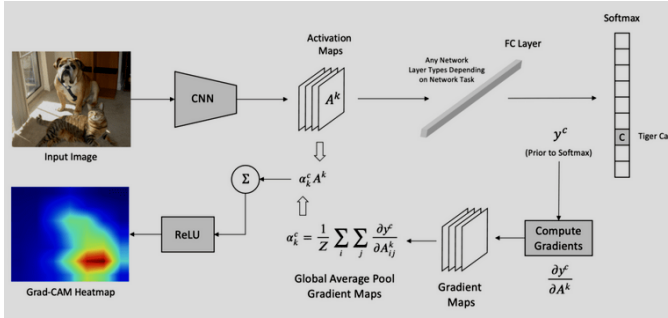


Fig. 1: Steps to compute Grad-CAM heatmaps from a trained CNN

In this context, Grad-CAM helps clinicians in medical imaging to understand where the model is ‘looking’ when making a classification. It can validate whether the model focuses on relevant anatomical structures, or highlight potential biases or errors.

III-B LIME (LOCAL INTERPRETABLE MODEL-AGNOSTIC EXPLANATIONS)

LIME is a model-agnostic technique that can be used to explain the predictions made by any supervised learning model, regardless of the underlying algorithm. As it treats the model as a black box, LIME can be used with virtually any classifier [15].

Rather than providing an overall understanding of the model across the entire dataset, LIME focuses on explaining the model’s predictions for individual instances by making small changes to the input data around a specific instance and observing how these changes affect the prediction.

LIME works by generating thousands of similar samples, obtaining their predictions using the original model, and assigning weights based on proximity to the original instance.

It then fits a simple, interpretable model — typically linear — to this surrogate dataset, in order to approximate the complex model’s local behavior around the instance of interest.

This local explanation reveals which input features had the greatest influence on the model’s decision for that particular instance, providing transparent, human-understandable insights regardless of the model’s complexity.

III-C SHAP (SHAPLEY ADDITIVE EXPLANATIONS)

SHAP is a model-agnostic method, meaning it is not restricted to a certain model type. It is also a local method, providing explanations for individual samples only. However, these individual explanations can also be used to obtain global interpretations [16]. It enables the fast computation of Shapley values and can be used to explain the prediction of an instance x by computing the contribution (Shapley value) of each feature to the prediction.

SHAP treats the prediction task as a game in which the features act as ‘players’ that contribute to the output. It calculates the average marginal contribution of each feature across all possible subsets of features, providing a theoretically sound and consistent attribution of feature importance.

Explanations of machine learning models based on Shapley values use game theory principles to distribute the contribution of each input feature to the model’s output fairly. In this analogy, features can be considered as players who can ‘participate’ or not in the game (model), which is equivalent to knowing or not knowing the value of these features [17].

SHAP offers a robust method of explaining model predictions both globally and locally, enabling users to grasp the influence of features in complex models and ensuring fair and interpretable results.

IV. DATA DESCRIPTION AND PREPROCESSING

In this section, we describe the PneumoniaMNIST dataset used in both our previous and current studies, and detail the preprocessing and augmentation steps applied. We also discuss the input resolutions evaluated—focusing on how intermediate higher resolutions (64x64 and 128x128) improve XAI interpretability.

IV-A DATASET OVERVIEW

PneumoniaMNIST is a pediatric chest X-ray subset of MedMNIST [1], [7], containing 5,856 grayscale images split into training, validation, and test sets. Table I summarizes its composition.

TABLE I: PneumoniaMNIST Composition and Class Distribution

Subset	Normal	Pneumonia	% of Total
Training (4,708)	1,214	3,494	80.4%
Validation (524)	135	389	8.9%
Testing (624)	234	390	10.6%

Pneumonia cases represent approximately 73% of all images, introducing class imbalance that can bias model perfor-

mance and must be accounted for during training, as we do via data augmentation.

Original images vary from 384x127 to 2,916x2,713 px. MedMNIST standardizes these via center-cropping (to the short side) and cubic-spline interpolation into 28x28, 64x64, 128x128, and 224x224 formats [1]. Our prior work used 28x28 by default; here, to enhance XAI map clarity, we also fine-tune models at 64x64 and 128x128, since intermediate resolutions have been shown to improve Grad-CAM heatmap granularity and SHAP attribution precision [18].

IV-B DATA VISUALIZATION

To illustrate the impact of image resolution on visual detail and its potential implications for XAI interpretability, we present sample images across the three resolutions used in our study. Higher resolutions preserve more anatomical detail, which is crucial for generating meaningful explanations from XAI techniques.

Figure 2 displays sample training images at 128x128 resolution, labeled Normal (0) and Pneumonia (1). At this higher resolution, radiographic patterns such as lung opacity, consolidation areas, and anatomical structures are more clearly discernible compared to the 28x28 baseline used in our previous work.

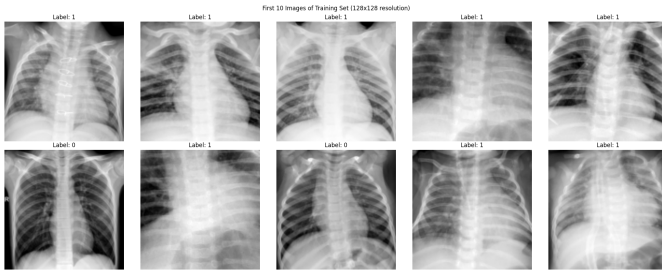


Fig. 2: Sample 128x128 PneumoniaMNIST Images (Normal vs. Pneumonia)

Figure 3 presents a direct comparison of the same chest X-ray images across 28x28, 64x64, 128x128, and 224x224 resolutions. This comparison demonstrates the progressive improvement in image quality and anatomical detail retention as resolution increases. Such detail enhancement is particularly important for XAI methods like Grad-CAM, which rely on spatial feature maps to generate meaningful heatmaps that can be interpreted by medical professionals.

While 224x224 resolution is available in the MedMNIST+ dataset, computational constraints in the Google Colab environment prevented us from training models at this resolution, as discussed in Section V. Therefore, our analysis focuses on the three computationally feasible resolutions that still demonstrate clear quality improvements for XAI applications.

IV-C PREPROCESSING STEPS

We apply a consistent preprocessing pipeline (Algorithm 1) before model input. Here, $h \times w$ denotes the chosen resolution

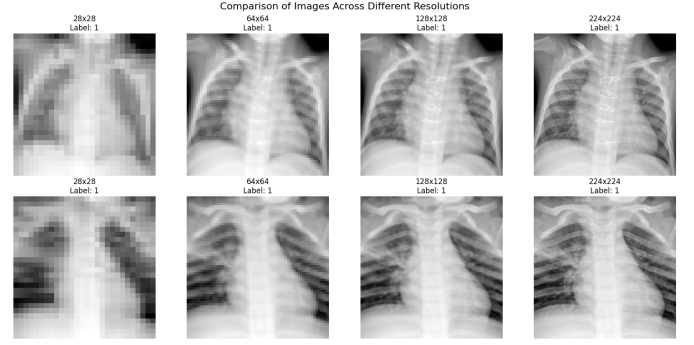


Fig. 3: Comparison of Same X-ray Images Across Different Resolutions (28x28, 64x64, 128x128)

(28, 64, 128, or 224).

Algorithm 1 Data Preprocessing Pipeline

```

1: procedure PREPROCESSDATA(images, labels, ( $h, w$ ))
2:   Reshape: images  $\rightarrow [n, h, w, 1]$   $\triangleright$  Add channel
3:   Normalize: images  $\leftarrow$  images / 255.0
4:   Encode: labels  $\rightarrow$  one-hot vectors (2 classes)
5:   return normalized images, encoded labels
6: end procedure

```

This ensures model compatibility across different CNN architectures while preserving data integrity.

IV-D DATA AUGMENTATION

To mitigate overfitting and promote generalization, we apply a standard augmentation pipeline to the training set only, as shown in Algorithm 2. This mirrors the techniques used in the first project, where data augmentation was crucial in improving the performance of the model. For this follow-up study, we are reusing the models that were trained on the augmented dataset because they achieved better results consistently across various architectures.

Algorithm 2 Data Augmentation Pipeline

```

1: procedure AUGMENTDATA(images, labels)
2:   pipeline  $\leftarrow$  SequentialAugmentation()
3:   pipeline.Add(RandomRotation(0.08))  $\triangleright \pm 15^\circ$  rotation
4:   pipeline.Add(RandomTranslation(0.1, 0.1))  $\triangleright 10\%$  shifts
5:   pipeline.Add(RandomZoom(0.1))  $\triangleright \pm 10\%$  zoom
6:   pipeline.Add(RandomContrast(0.1))  $\triangleright \pm 10\%$  contrast
7:   return pipeline(images), labels
8: end procedure

```

Figure 4 illustrates the visual effect of the augmentation process across different resolutions, demonstrating how the same augmentation parameters affect image quality and detail preservation at various scales.

To preserve evaluation integrity, no augmentation is applied to the validation or test sets, following best practices outlined in recent literature [10].

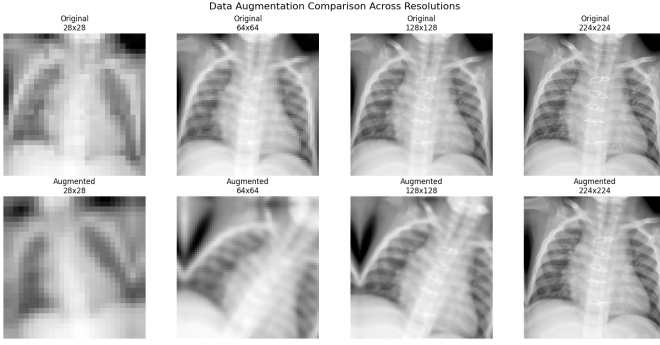


Fig. 4: Data Augmentation Effects Across Different Resolutions (28x28, 64x64, 128x128, 224x224)

V. METHODOLOGY

V-A OVERALL PIPELINE

Our methodology builds directly on the pipeline established in our previous work, with key extensions to support model fine-tuning at higher resolutions and the integration of explainable AI (XAI) techniques. The workflow consists of the following steps:

- 1) **Data preprocessing and augmentation** - Standardize images and apply augmentation techniques
- 2) **Model adaptation and fine-tuning** - Adapt architectures for different resolutions and fine-tune
- 3) **Classification and evaluation** - Assess performance using standard metrics
- 4) **Interpretability analysis using XAI** - Apply Grad-CAM, LIME, and SHAP techniques

V-B DATA PREPROCESSING AND AUGMENTATION

As described in Section IV, we use the PneumoniaMNIST dataset, standardizing all images to the target resolutions of 28x28, 64x64, and 128x128. Each image is normalized to the $[0, 1]$ range and labels are one-hot encoded. Data augmentation (including random rotations, translations, zoom, and contrast adjustments) is applied exclusively to the training set to mitigate overfitting and improve generalization. The preprocessing and augmentation steps are kept consistent across all experiments to ensure comparability.

V-C MODEL ADAPTATION AND FINE-TUNING

We evaluate four widely used CNN architectures: AlexNet, ResNet-18, ResNet-50, and DenseNet-169. Each model is adapted to accept the different input resolutions. For DenseNet-169, grayscale images are converted to three channels to match the expected input format. Transfer learning is leveraged by initializing DenseNet-169 with ImageNet weights, while the other models were trained from scratch and fine-tuned from previously trained weights at lower resolutions.

Fine-tuning is performed for each architecture at 64x64 and 128x128, building on the best hyperparameters identified in

our earlier 28x28 experiments. Table II summarizes the optimal hyperparameter configurations used for each architecture.

TABLE II: Optimal Hyperparameter Configurations by Architecture

Architecture	Augmentation	Batch Size	Dropout
AlexNet	Yes	96	0.5
ResNet-18	Yes	64	0.5
ResNet-50	Yes	32	0.3
DenseNet-169	No	96	0.5

The process involves freezing the convolutional base, training a new classification head, and then optionally unfreezing deeper layers for further adaptation. All models are trained using the same data splits for consistency.

It is important to note that, due to hardware limitations in the free Google Colab environment, training AlexNet at 224x224 resolution was not feasible as it exceeded available RAM. Similarly, later experiments indicated that ResNet-50 at 128x128 already approached memory limits, suggesting that 224x224 would also be impractical for this architecture. Therefore, our main comparative analysis focuses on 28x28, 64x64, and 128x128 resolutions.

V-D CLASSIFICATION AND EVALUATION

Model performance is assessed using the same metrics as in our previous work: accuracy, area under the ROC curve (AUC), precision, recall, F1-score, and confusion matrix. These metrics are computed on the held-out test set to ensure fair comparison. The class imbalance present in the dataset is addressed through data augmentation and by reporting metrics that are robust to imbalance, such as AUC and F1-score.

V-E EXPLAINABLE AI (XAI) INTEGRATION

The principal innovation in this phase is the systematic application of XAI techniques (Grad-CAM, LIME, and SHAP) to interpret model predictions, guided by established tutorials and documentation [12]–[14], ensuring best practices and reproducibility. For each architecture and resolution, we generate visual explanations for both correct and incorrect predictions, with a particular focus on false negatives and false positives.

V-E1 Implementation Details

Our XAI implementation follows a systematic approach designed to handle multiple models and resolutions while managing computational constraints. The analysis pipeline processes each trained model independently, loading the model and its corresponding test predictions, evaluating sample-level accuracy, and then applying each XAI technique based on predefined configuration settings.

To ensure robustness, each XAI method is executed with comprehensive error handling to prevent individual failures from interrupting the overall analysis workflow. This approach accounts for common issues such as architectural incompatibilities between XAI methods and specific model layers,

memory constraints, and varying input requirements across different techniques.

All analysis results, including predictions, probabilities, and XAI outputs are stored and serialized for subsequent analysis and visualization. Memory management is prioritized through explicit resource cleanup after each model analysis, including model deletion and backend session clearing, which proved essential when working within computational constraints.

The implementation generates summary statistics tracking the success rate of each XAI method across different model-resolution combinations, enabling systematic evaluation of method reliability and compatibility with different architectures.

a) Grad-CAM Implementation

Grad-CAM was implemented using the TF-Explain library. The `apply_gradcam_analysis()` function identifies the last convolutional layer of the model, scanning the layers in reverse order if none is explicitly provided. A heatmap is then generated for each input image by computing the gradients of the target class score with respect to the selected feature maps. These gradients are then globally averaged and weighted over the feature maps to produce class-discriminative localization maps. For each image and predicted class, the method returns a heatmap that highlights the most relevant spatial regions that contribute to the decision.

Special attention is given to exception handling to ensure that the analysis proceeds even if some images or models lack convolutional layers. In such cases, None values are returned for failed heatmaps.

b) LIME Implementation

The LIME method was implemented using the LIME image module. The `apply_lime_analysis()` function initializes a `LimeImageExplainer` and defines a prediction function that is compatible with grayscale input, ensuring the correct dimensionality. Each image is then perturbed by masking out super-pixels and a local linear model is trained on the perturbed samples to approximate the classifier’s behavior.

The method then calls `explain_instance()` for each image to extract the regions (superpixels) that contribute most to the model’s output. The returned explanation objects are stored for later visualization. Due to its computational cost, LIME is limited to a smaller subset of images, but the function is robust to shape mismatches and runtime failures.

c) SHAP Implementation

SHAP was implemented using the `Shap.DeepExplainer` library, which is compatible with Keras models. The `apply_shap_analysis()` function selects a subset of the test images to create the background dataset and calculates SHAP values for a limited number of samples. These values estimate the contribution of each input pixel to the model’s prediction.

The function returns class-wise SHAP values for all inputs, enabling subsequent visualization and comparison with other techniques. Due to SHAP’s high computational and memory requirements, the implementation restricts the analysis to a

predefined number of samples and incorporates error handling to prevent interruptions.

V-E2 Qualitative Analysis of Explanations

Given the lack of pixel-level pathology masks in PneumoniaMNIST, our interpretability analysis is primarily qualitative. We present side-by-side visualizations of XAI heatmaps overlaid on input images at different resolutions, allowing for direct comparison of explanation clarity and anatomical plausibility. This approach enables us to assess whether higher resolutions yield more focused or clinically meaningful explanations.

Ideally, expert radiologists would review these explanations to judge their clinical relevance; however, such expert evaluation was not feasible within our project time-frame. We acknowledge that expert assessment would provide a stronger validation of explanation quality.

We also strive for reproducibility by fixing random seeds and maintaining consistent preprocessing, but do not perform multiple training runs due to computational constraints.

V-E3 Potential Quantitative Metrics

If pixel-level ground-truth masks were available, we could employ quantitative localization metrics such as Intersection over Union (IoU), Dice coefficient, or the “pointing game” (measuring whether the most salient point in the heatmap falls within the annotated pathology region). Additionally, metrics such as explanation coverage (fraction of the image highlighted) and explanation consistency (stability across runs or similar images) could be considered. In our case, we limit our analysis to qualitative visual inspection and comparison across models and resolutions.

V-F EXPERIMENTAL SETUP

All experiments were conducted in a consistent environment to ensure fair comparison. Models were initially prototyped locally, but final training and evaluation were performed on Google Colab [8] due to memory constraints encountered locally (especially with DenseNet-169).

1) Computational Resources:

- Primary: Google Colab with NVIDIA T4 GPUs (16GB memory)
- Early: Local NVIDIA GPUs (1050Ti/1650Ti, 4GB memory)

2) Software Environment:

- TensorFlow 2.x, CUDA toolkit, Jupyter Notebook
- Utility libraries: NumPy, scikit-learn [9], Matplotlib, Seaborn

3) Experimental Design:

- Each architecture trained with identical procedures and fixed hyperparameters (see above)
- Training performed with or without augmentation as per best configuration
- Models evaluated using validation set during training; best model selected by validation AUC

- Final evaluation on held-out test set

Hyperparameters were standardized across experiments, with key parameters (batch size, dropout, augmentation) fixed per architecture as described above.

For medical image analysis, AUC and recall (sensitivity) were prioritized to minimize the risk of false negatives.

V-G EVALUATION METRICS

To ensure consistency and comparability with our previous work, we are reusing the evaluation metrics that were established in the first project. This systematic approach enables us to assess and compare the performance of models and explanation techniques objectively, under a shared evaluation framework. As before, we use the following set of classification metrics, which are computed using the scikit-learn implementation:

- 1) **Accuracy:** Measures the overall proportion of correct predictions across both classes.
- 2) **Area Under the ROC Curve (AUC):** Quantifies the model's ability to discriminate between normal and pneumonia cases across various classification thresholds, providing a threshold-independent performance measure.
- 3) **Precision and Recall:** Particularly important for medical diagnoses:
 - a) **Precision:** The proportion of true pneumonia cases among all cases predicted as pneumonia. Mathematically expressed as $\frac{TP}{TP+FP}$.
 - b) **Recall (Sensitivity):** The proportion of correctly identified pneumonia cases among all actual pneumonia cases. Mathematically expressed as $\frac{TP}{TP+FN}$.
- 4) **F1-Score:** The harmonic mean of precision and recall, providing a balanced metric particularly valuable for our potentially imbalanced dataset: $F1 = 2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$
- 5) **Confusion Matrix:** Visualizes the performance breakdown by showing true positives, false positives, true negatives, and false negatives, offering insights into specific classification errors.

VI. RESULTS AND DISCUSSION

This section presents the experimental results of our comparative study on CNN architectures for pneumonia detection at multiple resolutions, with a focus on model fine-tuning performance and explainable AI (XAI) analysis. We begin with quantitative performance metrics from the fine-tuning experiments, followed by an XAI interpretability analysis.

VI-A FINE-TUNING PERFORMANCE ANALYSIS

Building on our previous work at 28×28 resolution, we fine-tuned four CNN architectures (AlexNet, ResNet-18, ResNet-50, and DenseNet-169) to higher resolutions (64×64 and 128×128) to evaluate performance improvements and their impact on model interpretability.

VI-A1 Overall Model Performance

Table III summarizes the performance of all models across different resolutions, ranked by false negative count (the most critical metric for medical diagnosis). The hyperparameter configurations for each model are detailed in Table II. Models with resolution indicated in parentheses represent our baseline work at 28×28 pixels, while fine-tuned models were evaluated at 64×64 and 128×128 resolutions.

TABLE III: Model Performance Summary (Ranked by False Negative Count)

Model	FN	FP	AUC	Acc.	Prec.	Rec.	F1
ResNet-50 (28)	0	71	0.979	0.886	0.90	0.89	0.88
AlexNet (28)	0	90	0.970	0.856	0.88	0.86	0.85
DenseNet-169 (28)	1	119	0.951	0.808	0.85	0.81	0.79
ResNet-18 (28)	1	74	0.974	0.880	0.90	0.88	0.87
ResNet-18 (64)	6	51	0.973	0.909	0.91	0.91	0.91
ResNet-50 (128)	6	53	0.968	0.905	0.91	0.91	0.90
ResNet-18 (128)	6	52	0.970	0.907	0.91	0.91	0.90
ResNet-50 (64)	7	61	0.964	0.891	0.90	0.89	0.89
DenseNet-169 (128)	9	70	0.957	0.873	0.88	0.87	0.87
DenseNet-169 (64)	10	62	0.956	0.885	0.89	0.88	0.88
AlexNet (64)	13	75	0.926	0.859	0.87	0.86	0.85
AlexNet (128)	14	78	0.927	0.853	0.86	0.85	0.85

Main Findings:

- **Data augmentation at low resolution (28×28) is highly effective:** Both ResNet-50 and AlexNet achieved zero false negatives and strong AUC, showing that data diversity can outweigh resolution increases for critical errors.
- **Resolution vs. Sensitivity Trade-off:** Fine-tuning to higher resolutions improved overall accuracy (e.g., ResNet-18: 0.880 → 0.907-0.909) but increased false negatives, highlighting a tension between accuracy and sensitivity.
- **Architecture Patterns:** ResNet architectures were most stable across resolutions; DenseNet-169 improved most with fine-tuning; AlexNet's performance declined at higher resolutions.
- **Deployment Implications:** For maximum sensitivity, ResNet-50 (28) with augmentation is best; for balanced performance, ResNet-18 at 64×64 or 128×128 offers a good trade-off.

VI-A2 Resolution Impact Analysis

Figure 5 presents performance heatmaps showing the relationship between architecture, input resolution, and key metrics. The main patterns are:

- **False Negative Rate:** ResNet-50 and AlexNet achieve perfect sensitivity at 28×28, but this degrades at higher resolutions. ResNet-18 is more consistent across scales.
- **False Positive Rate:** DenseNet-169 has the highest FP rates, while ResNet architectures are more balanced.
- **AUC:** AUC does not always improve with resolution; ResNet-50 peaks at 28×28, suggesting low-res features plus augmentation are sufficient.
- **Accuracy:** ResNet-18's accuracy improves with resolution, but AlexNet and DenseNet-169 show inconsistent trends.

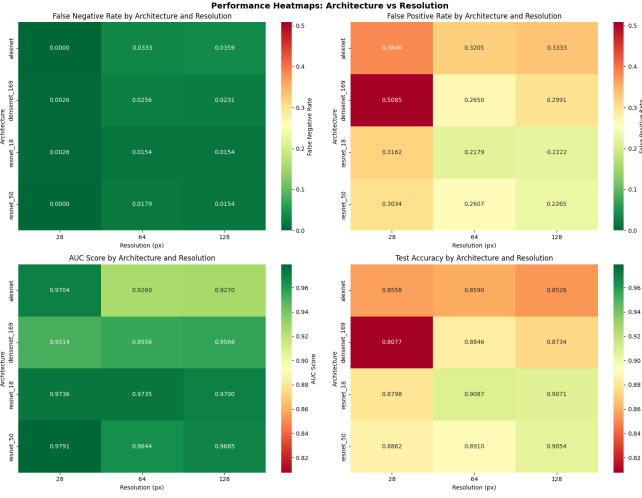


Fig. 5: Performance Heatmaps: Architecture vs Resolution Analysis for False Negative Rate, False Positive Rate, AUC Score, and Test Accuracy

Summary: Resolution scaling alone is not sufficient for optimal detection; data augmentation and regularization are often more impactful. Not all architectures benefit equally from higher resolution.



Fig. 6: Trade-off Analysis: False Negative vs False Positive Rate by Architecture and Resolution

The trade-off analysis (Fig. 6) shows that higher resolutions (128px) generally move models toward a better balance of FN and FP rates, especially for ResNet-18/50. However, the benefit is not uniform: AlexNet remains less optimal regardless of resolution.

The resolution improvement analysis (Fig. 7) shows that both 64×64 and 128×128 resolutions yield substantial gains in

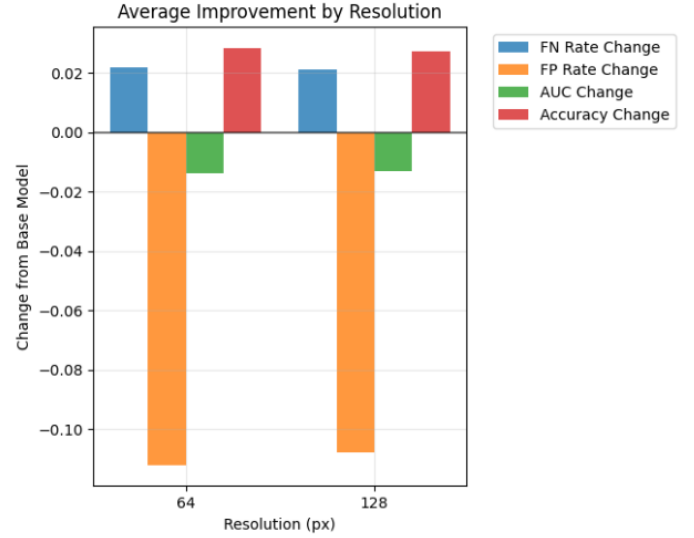


Fig. 7: Average Performance Improvement by Resolution

false positive (FP) reduction and small declines in false negative (FN) rates (due to higher being worse), when compared to the 28×28 baseline. The largest improvements are observed in FP reduction, while gains in sensitivity (FN) are more limited. Thus, higher resolution primarily helps reduce false positives relative to the lowest resolution, with incremental benefit for sensitivity. The effect size still varies by architecture.

VI-B EXPLAINABLE AI ANALYSIS

This section presents our primary contribution: a systematic analysis of how XAI techniques (Grad-CAM, LIME, and SHAP) perform across different model architectures and input resolutions for pneumonia detection.

VI-B1 XAI Technique Implementation and Comparison

We implemented three complementary XAI approaches to provide comprehensive model interpretability:

- 1) **Grad-CAM:** Applied to the final convolutional layer of each architecture to generate class activation heatmaps.
- 2) **LIME:** Configured with superpixel segmentation to identify relevant image regions.
- 3) **SHAP:** Using Deep Explainer to compute feature attribution scores.

To evaluate the practical robustness of each XAI technique, we tracked the success or failure of each method across all trained models and resolutions.

Table IV provides an overview of the methods that executed successfully, and Table V reports the number of sample-level explanations generated by each technique.

These results highlight implementation challenges: LIME frequently failed on ResNet variants due to input shape incompatibilities, while SHAP was memory-intensive and failed on

TABLE IV: Success of XAI method execution per model and resolution

Model	Resolution	Grad-CAM	LIME	SHAP
ResNet50	128px	✓	✗	✓
ResNet18	64px	✓	✗	✓
ResNet50	28px	✓	✗	✓
DenseNet169	28px	✓	✓	✗
DenseNet169	64px	✗	✓	✗
ResNet18	128px	✗	✗	✓
AlexNet	64px	✓	✓	✓
AlexNet	28px	✓	✓	✓
AlexNet	128px	✓	✓	✓
DenseNet169	128px	✓	✓	✗
ResNet50	64px	✗	✗	✗
ResNet18	28px	✓	✗	✓

TABLE V: Number of successful XAI results per method (out of 40 images)

Model	Resolution	Grad-CAM	LIME (5)	SHAP (1)
ResNet50	128px	40	0	1
ResNet18	64px	40	0	1
ResNet50	28px	40	0	1
DenseNet169	28px	40	5	0
DenseNet169	64px	0	5	0
ResNet18	128px	0	0	1
AlexNet	64px	40	5	1
AlexNet	28px	40	5	1
AlexNet	128px	40	5	1
DenseNet169	128px	40	5	0
ResNet50	64px	0	0	1
ResNet18	28px	40	0	1

DenseNet models. Grad-CAM demonstrated the greatest robustness overall, but also failed in models lacking compatible convolutional layers at certain resolutions.

To compare the quality of explanations across resolutions, we selected two representative architectures: AlexNet and ResNet50.

AlexNet is a relatively simple, older architecture that provides a baseline for understanding how resolution impacts the clarity of explanations in less complex models. In contrast, ResNet50, a deeper, more advanced convolutional network, shows how higher resolution benefits more sophisticated models with residual connections.

Figure 8 shows side-by-side comparisons of XAI heatmaps at 28×28 and 128×128 resolutions for both architectures. Improvements in spatial granularity, anatomical correspondence and explanation consistency are evident, emphasizing the importance of higher-resolution inputs for generating clinically meaningful interpretations. We can observe that higher resolution results in clearer and more anatomically relevant explanations.

VI-B2 Resolution Impact on Explanation Quality

Our analysis reveals significant improvements in explanation clarity and clinical relevance when transitioning from 28×28 to 128×128 resolution:

- **Spatial Granularity:** Higher resolutions enable more precise localization of pathological features, with 128×128 images showing clear delineation of opacity

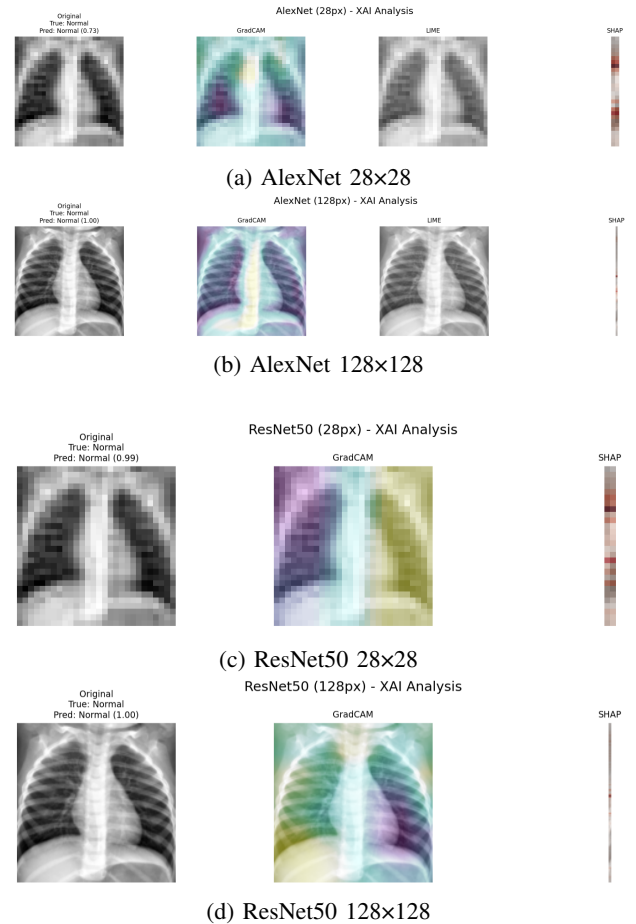


Fig. 8: Comparison of XAI heatmaps between 28×28 and 128×128 input resolutions for AlexNet (top row) and ResNet50 (bottom row)

patterns

- **Anatomical Correspondence:** XAI heatmaps at 128×128 resolution better align with expected anatomical structures (lung fields, costal margins)
- **Explanation Consistency:** Cross-technique agreement improves significantly at higher resolutions, suggesting more reliable interpretations

Figure 9 shows side-by-side comparisons of explanation heatmaps at 28×28, 64×64 and 128×128 resolutions for each architecture, illustrating these improvements. These visual enhancements highlight the importance of a sufficiently high input resolution when applying XAI methods in medical imaging contexts.

VI-B3 Architecture-Specific Interpretability Patterns

Different CNN architectures exhibit distinct explanation patterns that provide insights into their decision-making processes:

- **ResNet Architectures:** Generate focused, clinically plausible heatmaps that consistently highlight lung opacity regions

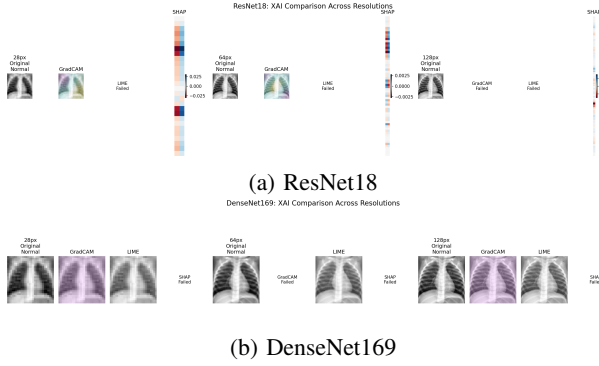


Fig. 9: Comparison of XAI explanation heatmaps at 28×28 and 128×128 input resolutions for different CNN architectures. Higher resolution consistently improves spatial detail, anatomical alignment, and explanation reliability.

- **DenseNet-169:** Produces more distributed attention patterns, potentially explaining its higher false positive rates
- **AlexNet:** Shows less consistent spatial attention, reflecting its simpler architectural design

These differences highlight the impact of architectural choices on the interpretability and reliability of explanations, emphasizing the importance of evaluating XAI outputs based on the underlying model.

In this analysis, we focus on 128×128 resolution heatmaps, as previous sections have demonstrated that this resolution produces clearer and more anatomically accurate explanations that are also clinically relevant, compared to lower resolutions of 28×28 and 64×64 .

Figure 10 shows architecture-specific XAI explanation patterns at 128×128 input resolution, revealing distinct attention distributions that reflect differences in model interpretability and decision behavior.

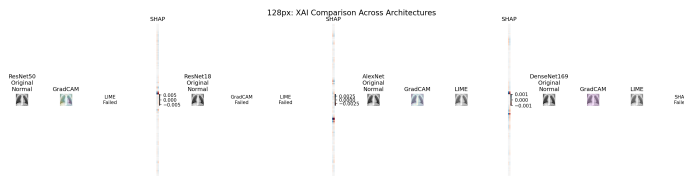


Fig. 10: Architecture-specific XAI explanation patterns at 128×128 input resolution for ResNet, DenseNet-169, and AlexNet

VII. DISCUSSION AND CONCLUSIONS

This study provides a comprehensive analysis of CNN architecture fine-tuning at multiple resolutions combined with XAI evaluation for pneumonia detection. Our findings reveal relationships between model architecture, input resolution, classification performance, and explanation quality that have important implications for clinical deployment.

VII-A FINE-TUNING EFFECTIVENESS ACROSS ARCHITECTURES

Our fine-tuning experiments reveal architecture-dependent responses to resolution scaling that provide insights into the fundamental capabilities of different CNN designs:

ResNet Architectures demonstrated the most consistent and stable performance across resolutions. ResNet-18 showed clear improvement with resolution scaling (accuracy: $0.880 \rightarrow 0.907$ - 0.909), while ResNet-50 achieved perfect sensitivity at 28×28 but experienced minor degradation at higher resolutions. The residual connections in these architectures enable effective gradient flow, facilitating the learning of fine-grained features at higher resolutions.

DenseNet-169 exhibited the largest performance improvement from fine-tuning, with accuracy increasing from 0.808 to 0.885 at higher resolutions. This substantial gain suggests that dense connections particularly benefit from additional spatial detail, likely due to enhanced feature reuse and propagation throughout the network.

AlexNet showed diminishing returns at higher resolutions, with performance plateauing or declining (accuracy: $0.856 \rightarrow 0.853$ - 0.859). This indicates that simpler architectures may lack the representational capacity to effectively take advantage additional spatial information.

These patterns demonstrate that resolution scaling alone is insufficient—architectural design fundamentally determines a model’s ability to benefit from higher-resolution inputs.

VII-B CRITICAL TRADE-OFFS IN MEDICAL AI DEPLOYMENT

Our analysis reveals a fundamental tension between sensitivity and overall performance that is crucial for clinical applications:

The Sensitivity Paradox: Models achieving zero false negatives at 28×28 resolution (ResNet-50 and AlexNet with augmentation) experienced degraded sensitivity when fine-tuned to higher resolutions. This finding challenges the assumption that higher resolution uniformly improves medical imaging performance.

Data Augmentation vs. Resolution: Our results demonstrate that data augmentation at low resolution can be more effective for critical error reduction than resolution increases alone. ResNet-50 with augmentation at 28×28 outperformed all higher-resolution variants in sensitivity.

Clinical Deployment Implications: For maximum sensitivity in pneumonia screening, ResNet-50 at 28×28 with augmentation remains optimal. However, for balanced diagnostic performance where both sensitivity and specificity matter, ResNet-18 at 64×64 or 128×128 provides superior overall metrics.

VII-C XAI RELIABILITY AND CLINICAL INTERPRETABILITY

Our systematic evaluation of XAI techniques reveals significant differences in robustness and practical applicability:

Method Reliability: Grad-CAM demonstrated superior robustness across architectures and resolutions, successfully generating explanations in 8 of 12 model-resolution combinations. LIME and SHAP faced substantial implementation challenges, with frequent failures due to architectural incompatibilities and computational constraints.

Resolution Impact on Explanation Quality: Higher resolutions (128×128) consistently produced more anatomically plausible and spatially precise explanations compared to 28×28 inputs. This improvement in interpretability quality represents a compelling reason to consider higher resolutions despite potential sensitivity trade-offs.

Architecture-Specific Patterns: ResNet models generated more focused and clinically plausible attention patterns, while DenseNet-169 produced more distributed attention that may explain its higher false positive rates or may even derive from the architecture itself, as all layers are connected. These findings suggest that architectural choice influences not only performance but also the quality and reliability of explanations.

VII-D CONTRIBUTIONS TO MEDICAL AI

This work advances the field of interpretable medical AI through the following contributions:

- 1) **Systematic Fine-Tuning Analysis:** Evaluation of CNN fine-tuning effects across multiple resolutions for pneumonia detection, revealing architecture-specific scaling behaviors.
- 2) **XAI Robustness Assessment:** Systematic evaluation of XAI technique reliability across medical imaging architectures, identifying Grad-CAM as the most robust approach for clinical applications.
- 3) **Resolution-Interpretability Relationship:** Demonstration that higher resolution improves explanation quality while potentially degrading sensitivity, establishing a new dimension for medical AI optimization.

VII-E LIMITATIONS AND FUTURE DIRECTIONS

Several limitations constrain the generalizability of our findings:

Clinical Validation Gap: The absence of expert radiologist evaluation represents the most significant limitation. Future work must prioritize clinical validation of both model predictions and XAI explanations to establish clinical utility.

Dataset and Scale Constraints: PneumoniaMNIST represents pediatric cases only and lacks pixel-level pathology annotations. Validation on adult populations and larger, more diverse datasets is essential for clinical translation.

Computational Limitations: Google Colab constraints prevented evaluation at 224×224 resolution and limited the scope of XAI analysis. Future studies with greater computational resources could explore additional resolution scales and more comprehensive XAI evaluations.

Methodological Extensions: Several areas warrant investigation, including: (1) quantitative interpretability metrics with ground-truth annotations, (2) expert-guided explanation quality assessment, (3) multi-institutional validation studies, and (4) investigation of XAI explanation consistency across similar cases.

VII-F CLINICAL RECOMMENDATIONS

Based on our findings, considering we are limited by data, hardware, and time constraints we can still extrapolate some recommendations:

- **For Pneumonia Screening:** ResNet-50 at 28×28 with data augmentation (zero false negatives, AUC: 0.979)
- **For Diagnostic Support:** ResNet-18 at 128×128 (balanced performance: AUC: 0.970, Accuracy: 0.907)
- **For Clinical Interpretability:** Grad-CAM with 128×128 input resolution for robust, anatomically meaningful explanations
- **For Development Workflows:** Prioritize data augmentation and architectural optimization before pursuing resolution increases

VII-G CONCLUSION

This study demonstrates that effective deployment of interpretable AI in medical imaging requires careful consideration of multiple interacting factors: architecture design, input resolution, training strategies, and interpretability requirements. Our findings challenge simplistic assumptions about resolution benefits while establishing practical guidelines for clinical AI development. The tension between sensitivity and interpretability quality revealed in our analysis represents a fundamental consideration for medical AI systems, emphasizing the need for application-specific optimization rather than one-size-fits-all solutions.

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